MIL-R-81158A (WP)
14 March 1960
Superseding
MIL-R-81158(Wep)
4 September 1964

MILITARY SPECIFICATION

REAR COVER ASSEMBLED UNIT

(FOR TORPEDO MARK 37 MODS 0 AND 1)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. Scorn

1*1 scope. This specification establishes the requirements for the procurement of a Rear Cover Assembled Unit, Bureau of Naval Weapons LD 620122, which provides the necessary connectors to adapt the acoustic panel to the wiring in the torpedo cabling.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

MILITARY

MIL-P-116	Preservation, Method of

STANDARDS

MILITARY

MIL-STD-15-1	Graphical Symbols for Electrical and Electronic Diagrams
MIL-STD-129	Marking for Shipment and Storage

FSC 1355

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DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 620122	Rear Cover, Assembled Unit
BUWEPS Dwg 1504214	Acoustic Panel
BUWEPS Dwg 1576589	Rear Cover Assembled Unit
BUWEPS Dwg 1576603	Rear Cover Schematic Diagram

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

WR-11	Design and Test of Packaging, Racking, Shipping, and Handling Equipment for Weapon System Components

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw Threads Standards for Federal Service

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

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3. REQUIREMENTS

3.1 Description. The rear cover is a flanged aluminum cover which contains the connectors necessary to adapt the wiring in the interconnection plate assembly to the wiring in the torpedo. The cover also provides a seal for the acoustic panel assembly.

3.2 General requirements

3.2.1 Contractor-furnished data. Data required by this specification, or by applicable documents referenced in Section 2, need not by furnished unless specified in the contract or order (see 6.2).

3.2.2 Preproduction samples. Unless otherwise specified in the contract Or order) preproduction samples of the rear cover shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the rear cover shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications, and publications listed in BUWEPS LD) 620122.

3.2.4 Surface treatment, construction, and dimensions. Surface treatment, construction, and dimensions for the rear cover shall be in accordance with applicable drawings, specifications, and publications listed in BUWEPS LD 620122.

3.2.5 Moisture, fungus, and corrosion. The rear cover shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in BUWEPS LD 620122.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in BUWEPS LD 620122.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

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3.2.8 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the materials to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.9 Requests for deviation. Each request that a deviation be authorized shall have been submitted to the procuring activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the procuring activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder), and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost or performance results therefrom.

3.2.10 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Symbols. Schematic diagram symbols shall be interpreted in accordance with MIL-STD-15-1.

3.4 Performance requirements and product characteristics. The rear cover shall meet the following performance requirements and product characteristics:

3.4.1 Continuity. All wires in the rear cover shall be electrically continuous, when tested as specified in 4.4.4.1.

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3.4.1.1 Continuity shall exist between each pin of J801 and the corresponding pin or pins of P801, as depicted on BUWEPS Dwg 1576603.

3.4,1.2 Continuity shall exist between each pin of J802 and the corresponding pin or pins of P802, as depicted on BUWEPS Dwg 1576603.

3.4.1.3 Continuity shall exist between each pin of J803 and the corresponding pin or pins of P803, as depicted on BUWEPS Dwg 1576603.

3.4.1.4 Continuity shall exist between each pin of J804 and the corresponding pin or pins of P804, as depicted on BUWEPS Dwg 1576603.

3.4.2 Dielectric. The rear cover shall withstand the specified potentials without breakdown, puncturing of insulation, or arcing when tested as specified in 4.4.4.2.

3.4.2.1 The dielectric withstanding voltage for connector 801 shall be 1000 volts rms at 60 cps for one minute, applied between each pin and all other pins and the connector shell, except for pin M. The dielectric withstanding voltage for pin M shall be 2400 volts rms at 60 cps for one minute applied between pin M and all other pins and the connector shell.

3.4.2.2 The dielectric withstanding voltage for connector 802 shall be 1000 volts rms at 60 cps for one minute applied between each pin and all other pins and the connector shell.

3.4.2.3 The dielectric withstanding voltage for connector 803 shall be 1000 volts rms at 60 cps for one minute applied between each pin and all other pins and the connector shell.

3.4.2.4 The dielectric withstanding voltage for connector 804 shall be 1000 volts rms at 60 cps for one minute applied between each pin and all other pins and the connector shell.

3.4.3 Insulation resistance. The rear cover shall meet the minimum insulation resistance requirements when tested as specified in 4.4.4.3.

3.4.3.1 With an applied voltage of 500 volts dc the insulation resistance shall be 30 megohms between each wire and all other wires and the connector shell. This test is to be performed on all connectors.

3.4.4 Direct current resistance. The rear cover shall meet the minimum direct-current resistance requirements when tested as specified in 4.4.4.4.

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3.4.4.1 The direct current resistance between the shell of P802 and the threaded hole marked GRD (located on the panel mounting flange) shall not exceed 0.20 ohm.

3.5 Environmental requirements

3.5.1 Vibration. The rear cover, mounted rigidly on a vibration device as specified in 4.5.2, shall be capable of meeting the following vibration requirements:

3.5.1.1 Vibration, transportation. The rear cover, preserved and packaged and packed as specified in the contract or order, shall be capable of withstanding the vibration tests specified in 4.5.2.1, and thereafter shall show no damage and shall meet all the performance and product requirements of this specification.

3.5.1.2 Vibration, nonoperating. The rear cover, less pack and packaging and self-mounted as specified in 4.5.2.2, shall be capable of withstanding the vibration tests specified in 4.5.2.2 and thereafter shall show no damage and shall meet all performance and product requirements of this specification.

3.5.1.3 Vibration, operating. The rear cover, self-mounted as specified in 4.5.2.3, shall be capable of withstanding the vibration tests specified in 4.5.2.3 and, during and after the tests, shall show no damage and shall meet all performance and product requirements of this specification.

3.5.2 Shock. The rear cover, mounted as specified in 4.5.3, shall be capable of meeting the following shock requirements:

3.5.2.1 Shock; transportation, storage and handling. The rear cover, preserved and packaged and packed as specified in the contract or order, shall be capable of withstanding the shock tests specified in 4.5.3.1, and thereafter shall show no damage and shall meet all performance and product requirements of this specification.

3.5.2.2 Shock, nonoperating. The rear cover, less pack and packaging and self-mounted as specified in 4.5.3.2, shall be capable of withstanding the shock tests specified in 4.5.3.2, and thereafter shall show no damage and shall meet-all performance and product requirements of this specification.

3.5.3 Temperature. The rear cover shall be capable of meeting the following temperature requirements:

3.5.3.1 Temperature, nonoperating condition. The rear cover shall be capable of withstanding temperatures between minus 65 degrees F and 160 degrees F for 76 hours as specified in 4.5.4.1, and thereafter shall show no damage and meet all performance and product requirements of this specification.

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3. 5.3.2 Temperature, operating condition. The rear cover shall show no damage and shall meet the performance and product requirements of this specification while being subjected to temperatures between zero degrees F and 175 degrees F as specified in 4.5.4.2.

3.5.4 Humidity. The rear cover shall show no corrosion or damage and shall meet all performance and product requirements of this specification after being subjected to humidity of 95 percent relative at temperatures between zero degrees F and 130 degrees F for the time periods as specified in 4.5.5.

3.5.5 Tropical environment, The rear cover shall show no fungus growth, corrosion, or other damage and shall meet the performance and product requirements of this specification after being subjected to the tropical conditions specified in 4.5.6.

3.5.6 Marine environment. The rear cover shall shov no corrosion or other damage and shall meet the performance and product requirements of this specification after being subjected to the simulated sea atmospheric condition specified in 4.5.7.

3.6 Cleanup. Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or exce s solder, metal chips and other foreign matter. Burrs and sharp edges as well as excessive rosin flash shall be removed.

3.7 Workmanship. The rear cover, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges.

3.7.1 The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may

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utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection of the rear cover shall be classified as follows:

a. Preproduction inspection (paragraph 4.5)
b. Acceptance inspection (paragraph 4.4)
c. Periodic production inspection (paragraph 4.5)

4.2 Sampling

4.2.1 Preproduction sampling. A preproduction sample of five rear covers manufactured in accordance with 3.2.2 shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the rear cover by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.2.2 Periodic production sampling. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.4. These samples shall be subjected to the tests of 4.5 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production, randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples from 3 consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of 50 units. Continue with Step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

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b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.2.3 Acceptance sampling. The tests and examinations detailed in 4.4 shall be conducted on each rear cover to be delivered; therefore, there shall be no sampling for acceptance. Defective units shall be rejected.

4.3 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.3.1 A continuity checker, NOD NMD/Y or equivalent.

4.3.2 A high potential test set capable of producing ac voltage of 60 cps, continuously variable from 0 to 3000 volts, and containing a sensing element to indicate insulation failure and a meter to indicate terminal voltage.

4.3.3 A megohmmeter capable of mess.uzing insulation resistance at 500 volts dc and including the range of 0-50 megohms with an accuracy of 2 percent.

4.3.4 An ohmmeter, AN/USM-21 or equivaelent.

4.4 Acceptance inspections

4.4.1 Each rear cover delivered under contract shall be subjected to the following tests and examinations. Failure to satisfactorily pass any test specified. therein shall be considered cause for rejection.

4.4.2 Test conditions. Unless otherwise specified, the rear cover shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18 degrees 0 to 35 degrees C).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

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4.4.3 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all acceptance tests. The test equipment shall be adequate in quality and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure

4.4.4.1 Continuity tests

4.4.4.1.1 For each combination of pins on J801 and P801, described in 3.4.1.1, connect one terminal of the continuity checker (4.3.1) to a pin of J801 and connect the other terminal of the checker to the corresponding pin of P801. In each case when the connections to the checker are completed the checker buzzer shall sound and the checker lamp shall light, indicating a closed circuit.

4.4.4.1.2 For each combination of pins on J802 and P802, described in 3.4.1.2, connect one terminal of the continuity checker to a pin of J802 and connect the other terminal of the checker to the corresponding pin of P802. In each case when the connections to the checker are completed the checker buzzer shall sound and the checker lamp shall light, indicating a closed circuit.

4.4.4.1.3 For each combination of pins on J803 and P803, described in 3.4.1.3, connect one terminal of the continuity checker to a pin of J803 and connect the other terminal of the checker to the corresponding pin in P803. In each case when the connections to the checker are completed the checker buzzer shall sound and the checker lamp shall light, indicating a closed circuit.

4.4.4.1.4 For each combination of pins on J804 and P804, described in 3.4.1.4, connect one terminal of the continuity checker to spin of J804 and connect the other terminal of the checker to the corresponding pin in P804. In each case when the connections to the checker are completed the checker buzzer shall sound and the checker lamp shall light, indicating a closed circuit.

4.4.4.2 Dielectric tests.

4.4.4.2.1 Connector 801 shall withstand a test potential of 2400 volts 60 cycle ac between pin M and all mutually insulated pins and between pin M and the connector shell for one minute minimum without breakdown.

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4.4.4.2.2 Connector 801 shall withstand a test potential of 1000 volts 60 cycle ac between pin L and all mutually insulated pins and between pin L and the connector shell for one minute minimum without breakdown.

4.4.4.2.3 With the exception of pins L and M, connector 801 shall withstand a test potential of 1000 volts 60 cycle ac between each pin and all mutually insulated pins and between each pin and the connector shell for one second without breakdown.

4.4.4.2.4 Connectors 802, 803, and 804 shall withstand a test potential of 1000 volts 60 cycle ac between each pin and all mutually insulated pins and between each pin and the connector shell for one second without breakdown.

4.4.4.3 Insulation resistance test. The insulation resistance between hipotted points shall be 30 megohms minimum at 500 volts dc.

4.4.4.4 Direct current resistance. Connect one lead of the ohmmeter (4.3.4) to the shell of P802 and the other lead of the ohmmeter to the threaded hole marked GRD (located on the panel mounting flange). The resistance indicated shall not exceed the minimum requirement specified in 3.4.4.1.

4.4.4.5 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking of the rear cover shall conform to Section 5 of this specification.

4.4.4.6 Visual and dimensional examination. Each rear cover shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the acceptance inspections detailed in 4.4, shall be subjected to the following tests and examinations.

4.5.1 Dielectric and insulation resistance

4.5.1.1 Connector 801 shall withstand a test potential of 2400 volts 60 cycle ac between pin M and all mutually insulated pins and between pin M and the connector shell for one minute minimum without breakdown.

4.5.1.2 With the exception of pin M, connector 801 shall withstand a test potential of 1000 volts 60 cycle ac between each pin and all mutually insulated pins and between each pin and the connector shell for one minute minimum without breakdown.

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4. 5.1.3 Connectors 802, 803, and 804 shall withstand a test potential of 1000 volts 60 cycle ac between each pin and all mutually insulated pins and between each pin and the connector shell for one minute minimum without breakdown.

4.5.1.4 The insulation resistance between hippotted points shall be 30 megohms minimum at 500 volts dc.

4.5.2 Vibration. The rear cover shall be mounted and vibration tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3. At the conclusion of each vibration test, the rear cover shall meet the performance and product requirements of this specification.

4.5.2.1 Vibration, transportation. The rear cover, packaged and packed as specified in Section 5 or the contract or purchase order, shall be vibrated in accordance with WR-11. Starting at 2 cps, gradually increase frequency until pack just begins to leave the table. At this point vibrate for a period of one hour.

4.5.2.2 Vibration, nonoperating. The rear cover, less pack and packaging, shall be mounted to a test fixture and vibrated according to the following schedule in each of the three directions shown in Figure 1.

Frequency Range	Amplitude	Minimum Test Duration		
		Room Temperature	-20°F	+130°F
10 to 30 cps	0.060 ± 0.006 in.	60 min.	15 min.	15 min.
31 to 60 cps	3g ± 0.3g	60 min.	15 min.	15 min.
61 to 100 cps	3g ± 0.3g	60 min.	15 min.	15 min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a 3 to 5 minute period for the duration of the test.

The notes in Figure 1 are applicable to this requirement.

4.5.2.3 Vibration, operating. The rear cover shall be mounted to a test fixture and vibrated according to the following schedule in each of the three directions shown in Figure 1.

Frequency Range	Amplitude	Minimum Test Duration Room Temperature
20 to 60 cps	lg ± 0.lg	15 min.
61 to 150 cps	lg ± 0.lg	15 min.

The frequency shall be continuously varied between each of the limits

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given at a uniform rate over a 3 to 5 minute period for the duration of the test.

The notes in Figure 1 are applicable to this requirement.

4.5.3 Shock. The rear cover shall be mounted and shock tested as specified in 4.5.3.1 and 4.5.3.2. At the conclusion of each shock test, the rear cover shall meet the performance and product requirements of this specification.

4.5.3.1 Shock; transportation, storage, and handling. The rear cover, packed and packaged as defined in Section 5, shall be subjected to the appropriate rough-handling tests specified in WR-11.

4.5.3.2 Shock, nonoperating. The rear cover, less pack and packaging, shall be mounted to a test fixture and subjected to three shock pulses of 60g plus or minus 6g peak magnitude, for not less than 8 ms duration, in each of the directions shown in Figure 1 (total 18 shock pulses).

The notes in Figure 1 are applicable to this requirement.

4.5.4 Temperature. The rear cover shall be subjected to the temperature tests specified in 4.5.4.1 and 4.5.4.2. At the conclusion of the tests, the rear cover shall meet the performance and product requirements of this specification.

4.5.4.1 Temperature, nonoperating. The rear cover shall be subjected to the following temperature cycle:

(a) Reduce the temperature to minus 65 degrees F plus or minus 5 degrees F (minus 53.89 degrees C plus or minus 2.78 degrees C) and maintain this temperature for a period of at least 36 hours.

(b) Increase the temperature to 160 degrees F plus or minus 5 degrees F (71.11 degrees C plus or minus 2.78 degrees C) and maintain this temperature for a period of at least 36 hours.

(c) Reduce the temperature to 68 degrees F plus or minus 5 degrees F (20 degrees C plus or minus 2.78 degrees C) and maintain this temperature for a period of at least four hours. Immediately following this temperature cycle, the rear cover shall meet the performance and product requirements of this specification.

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4.5.4.2 Temperature, operating condition. The rear cover shall be subjected to the following test procedure:

(a) Reduce the temperature to O degrees F plus or minus 5 degrees F (minus 17.78 degrees C plus or minus 2.78 degrees C) and after maintaining this temperature for a -period of four hours, perform the acceptance tests specified in 4.4.4.

(b) Increase the temperature to 40 degrees F plus or minus 5 degrees F (4.44 degrees C plus or minus 2.78 degrees C) in steps of 10 degrees F (5.56 degrees C) and stabilize for 30 minutes at each step. After the 30-minute stabilizing period at 40 degrees F (4.44 degrees C) perform the acceptance tests specified in 4.4.4.

(c) Increase the temperature to 70 degrees F plus or minus 5 degrees F (21.11 degrees C plus or minus 2.78 degrees C) in steps of 10 degrees F (5.56 degrees C) and stabilize for 30 minutes at each step. After the 30-minute stabilizing period at 70 degrees F, perform the acceptance tests specified in 4.4.4.

(d) Increase the temperature to 125 degrees F plus or minus 5 degrees F (51.7 degrees C plus or minus 2.78 degrees C) in steps of 10 degrees F (5.56 degrees C) and stabilize for 30 minutes at each step. After the 30-minute stabilizing period at 125 degrees F perform the acceptance tests specified in 4.4.4.

(e) Increase the temperature in steps of 10 degrees F (5.56 degrees C) stabilizing for 30 minutes at each step until 175 degrees F plus or minus 5 degrees F (76.67 degrees C plus or minus 2.78 degrees C) is reached. Stabilize for at least four hours and perform the acceptance tests specified in 4.4.4.

(f) Reduce the temperature to 80 degrees F plus or minus 5 degrees F (26.67 degrees C plus or minus 2.78 degrees C) in steps of 10 degrees F (5.56 degrees C) and stabilize for 30 minutes at each step.

(g) Reduce the temperature to 68 degrees F plus or minus 5 degrees F (20 degrees C plus or minus 2.78 degrees C) and stabilize for at least four hours. Immediately following this temperature cycle, the rear cover shall meet the performance and product requirements of this specification.

4.5.5 Humidity, nonoperating condition. The rear cover shall be subjected to the following temperature cycle:

(a) Increase the temperature to 130 degrees F plus or minus 5 degrees F (54.44 degrees C plus or minus 2.78 degrees C) and the humidity to 95 percent plus or minus 5 percent relative and maintain this condition for eight hours.

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(b) Reduce the temperature to 68 degrees F plus or minus 5 degrees F (20 degrees C plus or minus 2.78 degrees C) and maintain this condition for four hours.

(c) Reduce the temperature to 0 degree F plus or minus 5 degrees F (minus 17.78 degrees C plus or minus 2.78 degrees C) and maintain this condition for four hours.

(d) Increase temperature to 68 degrees F plus or minus 5 degrees F (20 degrees C plus or minus 2.78 degrees C) and maintain this condition for four hours. Immediately following this cycle, acceptance tests shall be conducted as specified in 4.4.4 and the rear cover shall snow no corrosion or damage and shall meet the Performance and other requirements of this specification.

4.5.6 Tropical environment. The rear cover shall be exposed to fungus tests as specified in "Tropical Envirornment in Section 4 of Specification MIL-T-18404(Wep). At the conclusion of this test the rear cover shall meet the performance and product requirements of this specification.

4.5.7 Marine environment. The rear cover shall be subjected to the simulated sea atmospher tests as specified in "Simulated Sea Atmosphere" in Section 4 of Specification MIL-T-18404(Wep). At the conclusion of this test the rear cover shall meet the performance and product requirements of this specification.

4.6 Failure of the rear cover to meet any of the requirements and tests of this specification shall be considered cause for rejection.

4.7 Functional tests. At the conclusion of the tests specified in 4.5.2.1 (Vibration, transportation) and 4.5.3.1 (Transportation, storage, and handling), functional and other tests, as explicable, shall be conducted on the rear cover to determine freedom from operational malfunction and compliance with all the requirements of Section 3.

4.8 Waiver of tests. Tests specified in 4.5.2.l (Vibration, transportation) and 4.5.3.1 (Transportation, storage, and handling) may be waived by the procuring agency for Level C packs only, provided the contractor guarantees the rear cover will be delivered to the using activity in such a condition that any rear cover selected shall be capable of meeting all requirements of Section 3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

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5.1.1 Level A

5.1.1.1 Cleaning, drying, and preservative application. Not aplicable.

5.1.1.2 Unit packaging. Unit packaging of the rear cover shall be in accordance with Method IIb of Specification MIL-P-116. The unit package shall contain one rear cover.

5.1.1.3 Intermediate package. Not applicable.

5.1.2 LevelB

5.1.2.1 Cleaning, drying, and preservative application. Not applicable.

5.1.2.2 Unit packaging. Unit packaging of the rear cover, shall be in accordance with Method 1C-2 of Specification MIL-P-116. The unit package shall contain one rear cover.

5.1.2.3 Intermediate package. Not applicable.

5.1.3 Level C

5.1.3.1 Cleaning, drying, and preservative application. Not applicable.

5.1.3.2 Unit packaging. Unit packaging of the rear cover shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one rear cover.

5.1.3.3 Intermediate package. Not applicable.

5.2 Packing

5.2.1 Level A

5.2.1.1 Exterior containers. Rear covers packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601, overseas type, or PPP-B-621, class 2 (weight limit not to exceed 150 lbs) or in a fiberboard box in accordance with Specification PPP-B-636, class 2 (weight limit not to exceed 70 lbs).

5.2.2 Level B. Same as Level A except that boxes shall be of the domestic type.

5.2.3 Level C. Rear covers packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated weight Classification Rules or other common carrier regulations applicable to the mode of transportation.

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5.3 Cushioning. The *unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the rear cover will meet all the requirements of Section 3 after being tested in accordance with 4.5.2.1 and 4.5.3.1. At the time of packaging or packing, the moisture content of the cushioning material shall not exceed five percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. The rear cover is intended for use in Torpedo Mk 37 Mods 0 and 1.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable drawings and other documents.

(c) Required level of preservation and packaging.

(d) Required level of packing.

(e) Assigned activity for preproduction and periodic tests.

(f) Any exceptions to this specification.

(g) All additional samples required shall be provided and inspected at the expense of the supplier, when such. additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

(h) Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Bureau of Naval Weapons when testing is accomplished at other than a Government activity.

6.2.1 The attention of the contracting officer is invited to the options in 4.8.

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6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

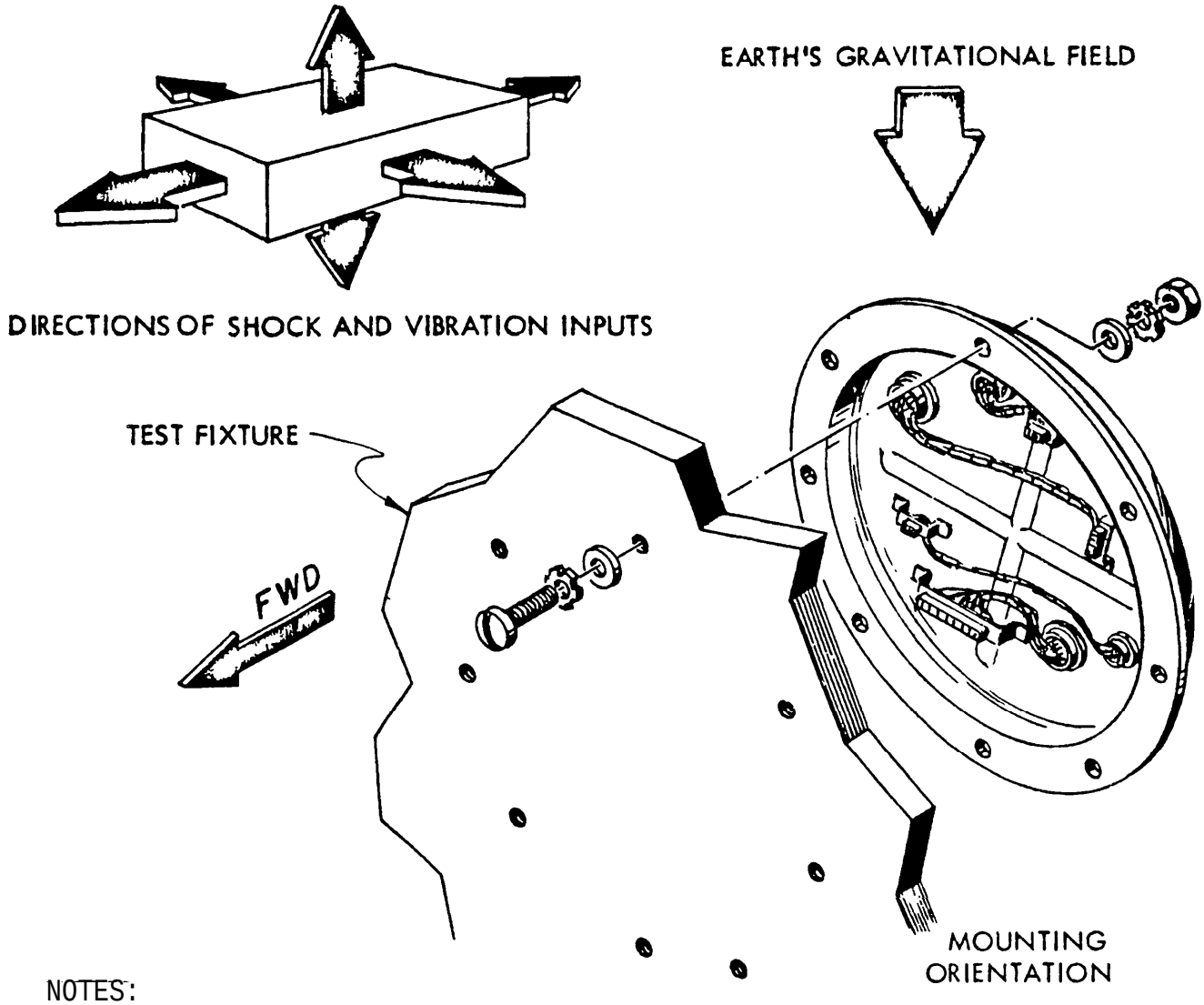
Level B. This-level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for six months or less or in domestic locations for an indefinite period.

Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. The contract or order shall include a requirement for a guarantee stating that any rear cover which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

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FIGURE 1. REAR COVER
MOUNTING REQUIREMENTS



NOTES:

1. FOR VIBRATION, ORIENTATION OF TEST ITEM WITH RESPECT TO EARTH'S GRAVITATIONAL FIELD SHALL BE MAINTAINED FOR ALL TEST DIRECTIONS.

2. FOR SHOCK, ORIENTATION OF THE TEST ITEM CAN BE CHANGED AS NEEDED TO ACCOMPLISH ALL DIRECTIONS OF TEST.

3. FOR BOTH SHOCK AND VIBRATION, THE TEST ITEM SHALL BE SECURED TO THE TEST FIXTURE AS SHOWN.

4. THE TEST FIXTURE SHALL NOT ALTER THE INPUTS FROM SHOCK OR VIBRATION EXCITER TO THE MOUNTING POINTS OF THE TEST ITEM.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-81158A(WP) Rear Cover Assembled Unit (For Torpedo Mark 37 Mods 0 and 1)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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